

Van der Pauw Resistivity Measurements of Germanium sample

Aim

To determine the electrical resistivity (ρ) of a germanium sample using the van der Pauw measurement method.

Apparatus

- 4200A-SCS Parameter Analyzer
- Four Source Measure Unit (SMU) instruments
- Four preamplifiers (for high resistance material testing, if applicable)
- Low noise shielded triax cables
- Germanium sample



Figure : Germanium sample (n-type , 0.5mm thickness) on puck with Van der Pauw contacts

Theory

The van der Pauw method is used to determine the resistivity of a flat, arbitrarily shaped sample of uniform thickness by applying current and measuring voltage through four small contacts located on the sample's circumference. Because the specific geometric spacing of the contacts does not impact the result, this method is highly effective for thin films.

Resistivity is derived from a total of eight voltage measurements taken around the periphery of the sample. These measurements are used to calculate two volume resistivities, ρ_A and ρ_B :

$$\rho_A = \pi / \ln 2 \cdot f_A t_s \cdot (V_1 - V_2 + V_3 - V_4) / 4I$$
$$\rho_B = \pi / \ln 2 \cdot f_B t_s \cdot (V_5 - V_6 + V_7 - V_8) / 4I$$

Where:

- t_s is the germanium sample thickness in cm.
- I is the current applied through the sample in amperes.
- V_1 through V_8 represent the specific voltages measured by the voltmeters.
- f_A and f_B are geometrical factors based on the sample's symmetry ($f_A = f_B = 1$ for perfect symmetry).

Once ρ_A and ρ_B are calculated, the average volume resistivity (ρ_{AVG}) is determined using the equation:

$$\rho_{AVG} = (\rho_A + \rho_B) / 2$$

Procedure

1. Sample Connection: Connect each of the four terminals on the germanium sample to one of the four SMU instruments using shielded triax cables to minimize electrostatic interference. Ensure the inner shield of the triax connector is used as a guard to reduce leakage current.
2. Initial Configuration: Configure the instruments for the first test setup. Set one SMU as the current source (using a two-point current list sweep), two SMUs as high-impedance voltmeters (testing with 0 A current), and the fourth SMU as common.
3. Current Adjustment: Adjust the source current magnitude based on the film's expected resistance. The applied current should be kept low enough that the resulting voltage difference does not exceed approximately 25 mV to prevent carrier injection from skewing the measurement.
4. Measurement & Offset Correction: Source both the positive and negative values of the test current and measure the corresponding voltages. Average these voltage measurements for both polarities to

successfully correct for internal voltage offsets or thermoelectric voltages in the circuit.

5. Permutation: Rotate the SMU measurement setup around the sample (changing which terminals act as the source, voltmeters, and common) and repeat the procedure until four complete test configurations have been executed.
6. Data Processing: Extract the averaged voltage differences. When plugging the measured voltage outputs into the final calculations, verify that all values are explicitly recorded in Volts (rather than millivolts) to ensure the standard units are maintained across the equations.
7. Calculation: Use the recorded test current, the measured voltage sums, and the known thickness of the germanium film to calculate ρ_A and ρ_B , and subsequently find the overall ρ_{AVG} .

Observations & Calculations

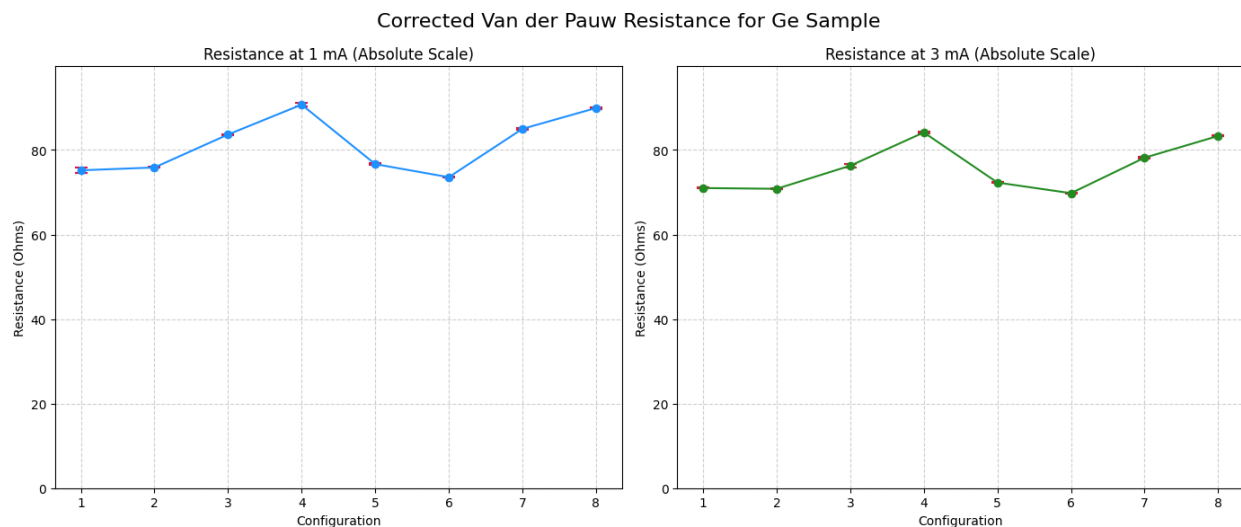


Figure : Calculated Mean Resistances for each configuration

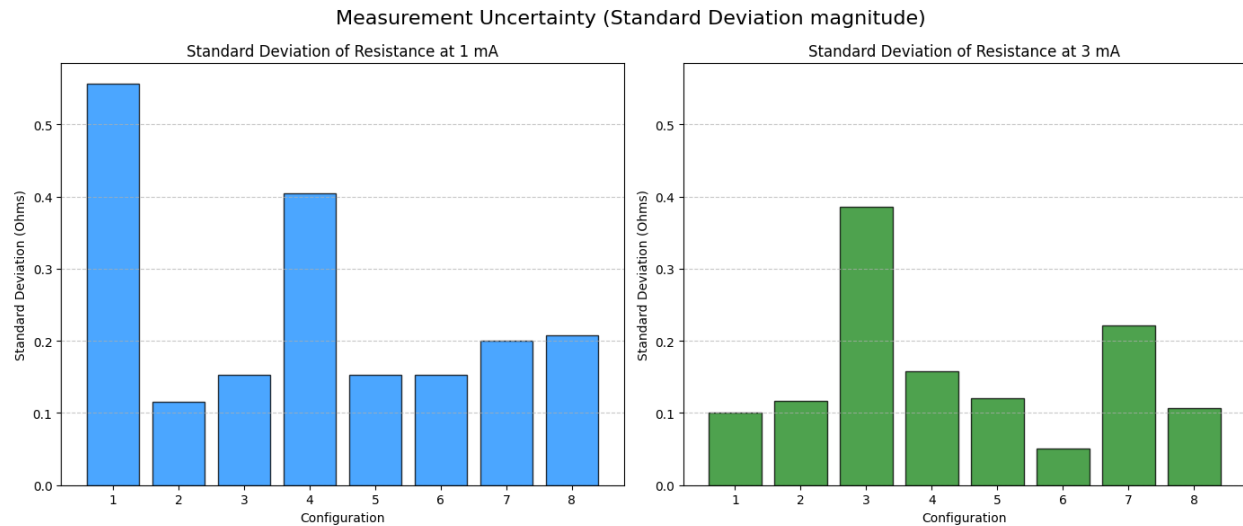


Figure : Calculated Error in Resistances for each configuration

Raw voltage measurements for $I = 1 \text{ mA}$

Measurement	Trial 1 (V)	Trial 2 (V)	Trial 3 (V)
V1	0.747	0.751	0.758
V2	0.758	0.760	0.758
V3	0.837	0.838	0.835
V4	0.903	0.908	0.911
V5	0.765	0.767	0.768
V6	0.734	0.736	0.737
V7	0.848	0.850	0.852
V8	0.897	0.900	0.901

Raw voltage measurements for $I = 3 \text{ mA}$

Measurement	Trial 1 (V)	Trial 2 (V)	Trial 3 (V)
V1	2.127	2.130	2.133
V2	2.121	2.125	2.128

Measurement	Trial 1 (V)	Trial 2 (V)	Trial 3 (V)
V3	2.278	2.287	2.301
V4	2.523	2.521	2.530
V5	2.165	2.170	2.172
V6	2.094	2.095	2.092
V7	2.341	2.342	2.353
V8	2.502	2.496	2.501

Results

Results at 1 mA Source Current

- Average RA Axis Resistance: $75.325 \pm 0.088 \, \Omega$
- Average RB Axis Resistance: $87.333 \pm 0.075 \, \Omega$
- Calculated Sheet Resistance (R_s): $368.140 \pm 0.366 \, \Omega/\text{sq}$
- Calculated Volume Resistivity (ρ): $18.4070 \pm 0.0183 \, \Omega \cdot \text{cm}$

Results at 3 mA Source Current

- Average RA Axis Resistance: $71.492 \pm 0.024 \, \Omega$
- Average RB Axis Resistance: $79.542 \pm 0.038 \, \Omega$
- Calculated Sheet Resistance (R_s): $341.491 \pm 0.141 \, \Omega/\text{sq}$
- Calculated Volume Resistivity (ρ): $17.0745 \pm 0.0071 \, \Omega \cdot \text{cm}$

Precautions

- Contact Placement: Ensure the four ohmic contacts are as small as possible and placed exactly on the edges or corners of the sample.
- Current Limits: Keep the applied test current low enough that the measured voltage difference does not exceed approximately 25 mV. Higher currents can cause sample heating and carrier injection, which skew results.
- Offset Cancellation: Always measure the voltage using both positive and negative current polarities, then average the readings to cancel out internal thermoelectric voltage offsets.
- Shielding and Guarding: Use low-noise shielded triax cables and ensure the circuit is properly guarded to minimize electrostatic interference and leakage current.
- Light Isolation: Keep the sample in a completely dark environment during testing to prevent light from generating false photoconductive currents.